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```
%%Case Study 1: K-Means Clustering
% Name: Derek Nester
```

```
clear;
%close all;
```

### --- Initialize Data Set ---

```
fprintf('Loading training data...\n');
train = csvread('mnist_train_1500.csv');
trainsetlabels = train(:,785);
train = train(:,1:784);
train(:,785) = zeros(1500,1);
```

*Loading training data...*

### --- K-Means Algorithm Setup ---

```
k = 18; % SET THIS VALUE from the result of your BestK.m script
max_iter = 50;
fprintf('Starting k-means clustering with k = %d...\n', k);
```

```
% Initialize the centroids
centroids = initialize_centroids(train, k);
cost_iteration = zeros(max_iter, 1);
```

*Starting k-means clustering with k = 18...*

### --- Run K-Means Clustering ---

```
for iter = 1:max_iter
    current_cost = 0;
    % Assignment Step
    for i = 1:size(train, 1)
        [idx, dist] = assign_vector_to_centroid(train(i, 1:784), centroids);
        train(i, 785) = idx;
        current_cost = current_cost + dist^2;
    end
```

---

```

cost_iteration(iter) = current_cost;
fprintf('Iteration %d, Cost: %f\n', iter, current_cost);

% Update Step
centroids = update_Centroids(train, k);
end
fprintf('Clustering complete.\n');

Iteration 1, Cost: 6112859360.000000
Iteration 2, Cost: 3719672352.387302
Iteration 3, Cost: 3618112594.400471
Iteration 4, Cost: 3568847969.417141
Iteration 5, Cost: 3532186962.165457
Iteration 6, Cost: 3508834424.819142
Iteration 7, Cost: 3492997104.138372
Iteration 8, Cost: 3479427327.648709
Iteration 9, Cost: 3468139702.822229
Iteration 10, Cost: 3458932354.389961
Iteration 11, Cost: 3452711612.217251
Iteration 12, Cost: 3447431315.816057
Iteration 13, Cost: 3442780766.653005
Iteration 14, Cost: 3439962489.014465
Iteration 15, Cost: 3438723966.167984
Iteration 16, Cost: 3437904424.635896
Iteration 17, Cost: 3437158953.212049
Iteration 18, Cost: 3436014462.695443
Iteration 19, Cost: 3435033352.303622
Iteration 20, Cost: 3434256034.645561
Iteration 21, Cost: 3433747633.460319
Iteration 22, Cost: 3433160317.887812
Iteration 23, Cost: 3432837520.306648
Iteration 24, Cost: 3432682439.783457
Iteration 25, Cost: 3432605363.651381
Iteration 26, Cost: 3432458283.259130
Iteration 27, Cost: 3432281840.059775
Iteration 28, Cost: 3432130448.548777
Iteration 29, Cost: 3432056393.484407
Iteration 30, Cost: 3431885541.234366
Iteration 31, Cost: 3431717921.961974
Iteration 32, Cost: 3431589290.304242
Iteration 33, Cost: 3431589290.304242
Iteration 34, Cost: 3431589290.304242
Iteration 35, Cost: 3431589290.304242
Iteration 36, Cost: 3431589290.304242
Iteration 37, Cost: 3431589290.304242
Iteration 38, Cost: 3431589290.304242
Iteration 39, Cost: 3431589290.304242
Iteration 40, Cost: 3431589290.304242
Iteration 41, Cost: 3431589290.304242
Iteration 42, Cost: 3431589290.304242
Iteration 43, Cost: 3431589290.304242
Iteration 44, Cost: 3431589290.304242
Iteration 45, Cost: 3431589290.304242
Iteration 46, Cost: 3431589290.304242

```

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```
Iteration 47, Cost: 3431589290.304242
Iteration 48, Cost: 3431589290.304242
Iteration 49, Cost: 3431589290.304242
Iteration 50, Cost: 3431589290.304242
Clustering complete.
```

## --- MNIST Example Image ---

```
% Get the pixel data for the first image (row 1, columns 2-end)
first_image_vector = train(10, 1:784);

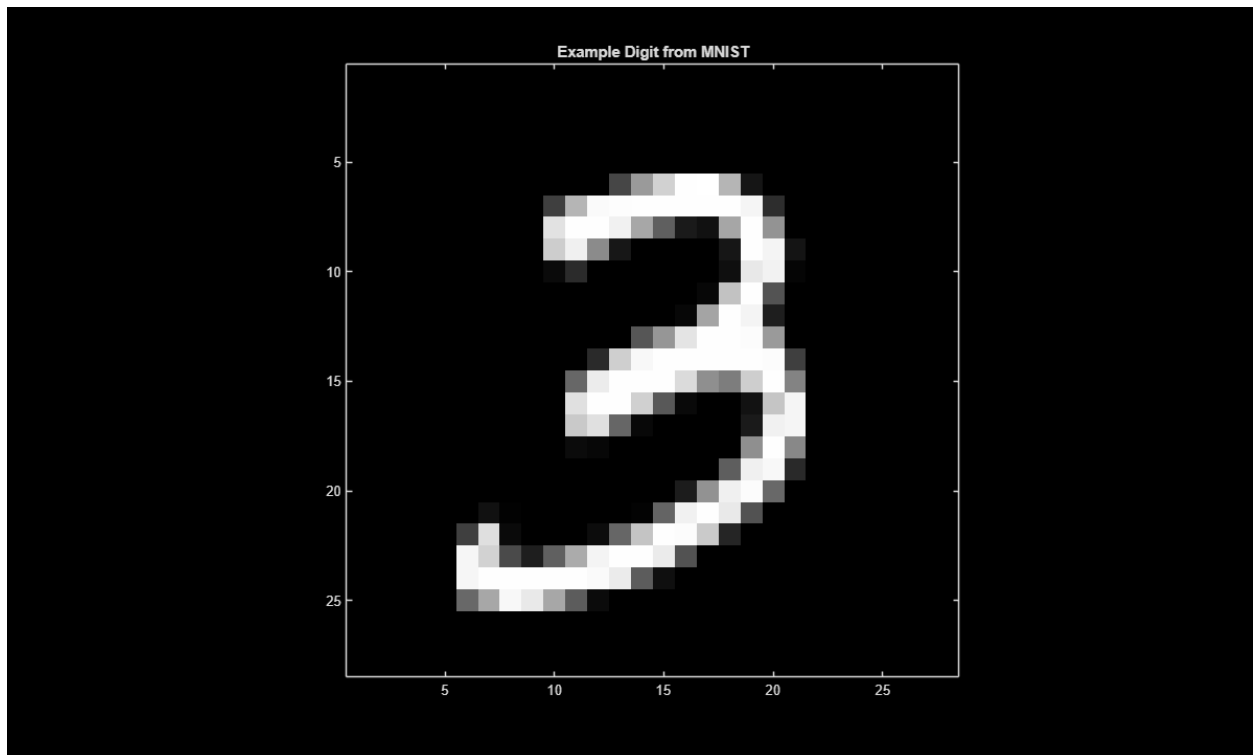
% Reshape the 784-vector into a 28x28 matrix and transpose it
image_matrix = reshape(first_image_vector, [28 28])';

figure;
imagesc(image_matrix);
colormap(gray);
axis square;
title('Example Digit from MNIST');

% Save the figure
saveas(gcf, 'mnist_example.png');

fprintf('Saved mnist_example.png\n');

Saved mnist_example.png
```



---

## --- Calculate Final Model Parameters and Save

---

```
fprintf('Calculating final model parameters...\n');

% Calculate the label for each centroid based on the training data
centroid_labels = zeros(k, 1);
for i = 1:k
    indices_in_cluster = find(train(:, 785) == i);
    if ~isempty(indices_in_cluster)
        labels_in_cluster = trainsetlabels(indices_in_cluster);
        centroid_labels(i) = mode(labels_in_cluster);
    else
        centroid_labels(i) = -1; % Handle empty clusters
    end
end

% Calculate distance thresholds for outlier detection from the training data
distance_thresholds = zeros(k, 1);
for i = 1:k
    indices_in_cluster = find(train(:, 785) == i);
    if ~isempty(indices_in_cluster)
        distances = zeros(length(indices_in_cluster), 1);
        for j = 1:length(indices_in_cluster)
            point_idx = indices_in_cluster(j);
            distances(j) = norm(train(point_idx, 1:784) - centroids(i, 1:784));
        end
        distance_thresholds(i) = mean(distances) + 2 * std(distances);
    else
        distance_thresholds(i) = inf; % No threshold for empty clusters
    end
end

% Save the final centroids, their labels, and the outlier thresholds
save('classifierdata.mat', 'centroids', 'centroid_labels', 'distance_thresholds');
fprintf('Classifier data saved to classifierdata.mat\n');

Calculating final model parameters...
Classifier data saved to classifierdata.mat
```

## --- Generate Required Plots for Clustering Script ---

```
% Plot 1: K-Means Cost vs. Iteration
figure;
plot(1:max_iter, cost_iteration, '-o', 'LineWidth', 2);
title('K-Means Cost Function vs. Iteration Number');
xlabel('Iteration Number');
```

---

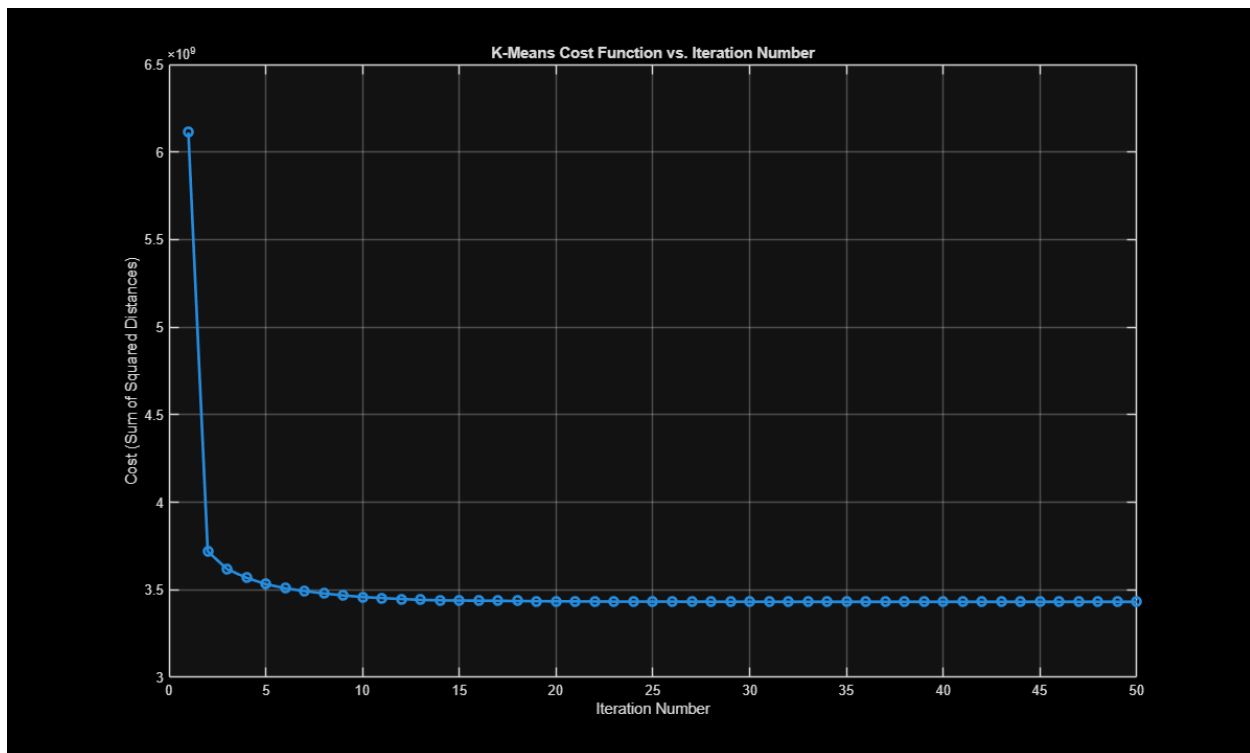
```

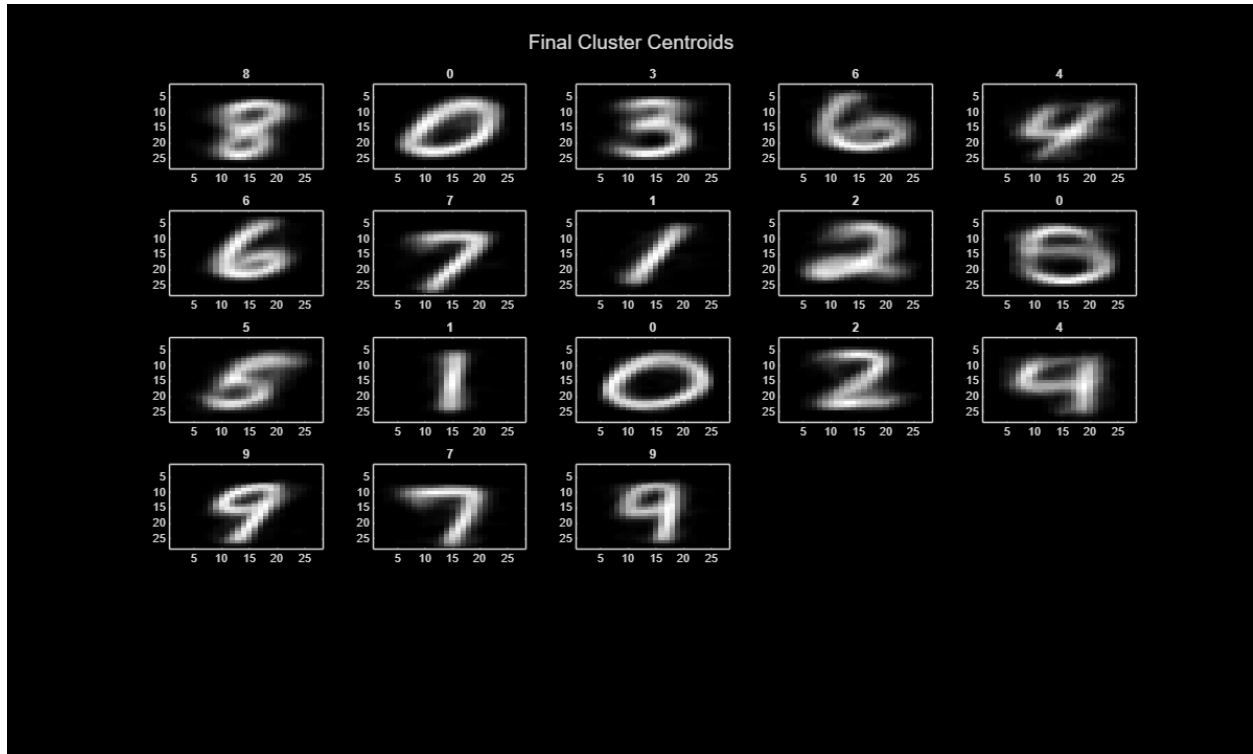
ylabel('Cost (Sum of Squared Distances)');
grid on;
saveas(gcf, 'cost_vs_iteration.png');

% Plot 2: Final Centroids
figure;
colormap('gray');
plotsize = ceil(sqrt(k));
for ind = 1:k
    centr = centroids(ind, 1:784);
    subplot(plotsize, plotsize, ind);
    imagesc(reshape(centr, [28 28]));

    title(centroid_labels(ind));
end
sgtitle('Final Cluster Centroids');
saveas(gcf, 'final_centroids.png');

```





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